

Rapid PROTOTYPING AND DEVELOPMENT Using Python Based Platforms

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Zerynth

Zerynth, which was formerly called VIPER (Viper Is Python Embedded in Real-Time), is a solution for making designs interactive and connected to the Internet of Things (IoT) easily. The package offers a set of software and hardware tools to seamlessly implement embedded design ideas, without worrying about actually putting together stuff using soldering to make designs work.

The suite is open source and released under general-public license version 3 (GPL3), Apache2 and Massachusetts Institute of Technology (MIT) licences, depending on the libraries and codes put to use. For commercial usage, their respective websites offer a separate package along with added support. Built to be cross-platform, Zerynth is a development suite for high-level design of interactive objects, artistic installations and Internet/Cloud-connected devices.

MyHDL

MyHDL is an open source platform developed by Jan Decaluwe for using Python, a general-purpose high-level language for hardware design. A designer who wishes to use this software can benefit from the power of Python language as well as the merits of free, open source software. Additionally, this platform helps in converting the design to Verilog and VHDL while supporting simulation and verification functions. This is a perfect hardware design tool to experiment with ideas, and integrates the use of dynamic language directly for hard-

Some popular resources

Programming Without Coding Technology (PWCT). The greatest take-away from this software is that anybody could become a programmer. Powered with general-purpose visual programming capabilities, a novice or an expert programmer can make the best use of this tool while using the programming concepts of data structure, control structure, programming paradigm and so on. An expert can use it to develop large complex software. With support for languages like Harbour, Supernova, C, Python and C#.NET, a user can view or edit the generated source code.

Tiny bootloader. This is a short program used to load firmware to the microcontroller without any programmer device like flash or volatile memory like RAM. It jumps to the desired program from where it takes care of execution. This program is the first thing to run, can load other applications into specific places in memory and is provided to the serial interface. Tiny bootloader can be specifically used to program Microchip, Atmel, NXP, TI and 8051 devices.

ware design, making way for unified algorithm and implementation. The software can be installed on any platform that supports Python.

CadSoft Eagle

Eagle PCB design software is a popular tool of choice for many design engineers and academia across the world. It comprises features such as schematic capture editor, PDB layout editor and auto router, to name a few. It is important that the software runs in Run as Freeware option to access the light edition for free and non-commercial or evaluation usage.

Rapid SCADA

Rapid SCADA is free, open source, full-featured supervisory control and data acquisition (SCADA) software. This software allows you to create automated systems around industrial automation systems, home automation systems, energy accounting systems, security fire alarm systems and other similar systems that contain controllers and relays.

EasyEclipse

EasyEclipse utility software is a pre-packaged release of Eclipse that provides a fully-functional integrated development environment for any specific type of development. This DVD edition carries development environment specifically for Python programming, and all you need to do is to start developing the code, without any additional downloads. It comes with many goodies such as code completion, syntax highlighting, syntax analysis, refactor, debug and others.

Fritzing

Fritzing revolves around building a creative arena for you to document and share your prototypes. This open source hardware initiative makes information related to electronics accessible. With this edition of Fritzing, critical bug fixes for PCB traces have been resolved. Some popular parts such as ESP8266 Thing development board, Arduino Pro Mini 3.3V and 5V, Arduino Mega Pro 3.3V, CAN-Bus Shield and so on are brought into this version. **EFY**

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